

Feature Importance

It refers to a technique that assigns scores to input Features, indicating their relative contribution to a ML model's predictions.

It aids in identifying the most impactful Features for model performance, interpretation and reducing overfitting.

Common methods include tree-based impurity reduction, permutation importance, and linear model.

Key Methods to calculate importance

1) Tree-Based (Impurity-Based)

Models like Decision Tree and Random

Forests Calculate importance by the total reduction in a criterion

(eg. Gini Impurity or Entropy) brought by that Feature, often called as Gini Importance.

Permutation Feature Importance

This method measures the increase in the model's prediction error after permuting (shuffling) a Feature's values, which breaks the relationship between the Feature & the target.

It is often referred because it is not biased toward high-cardinality Features. (High-cardinality = Column

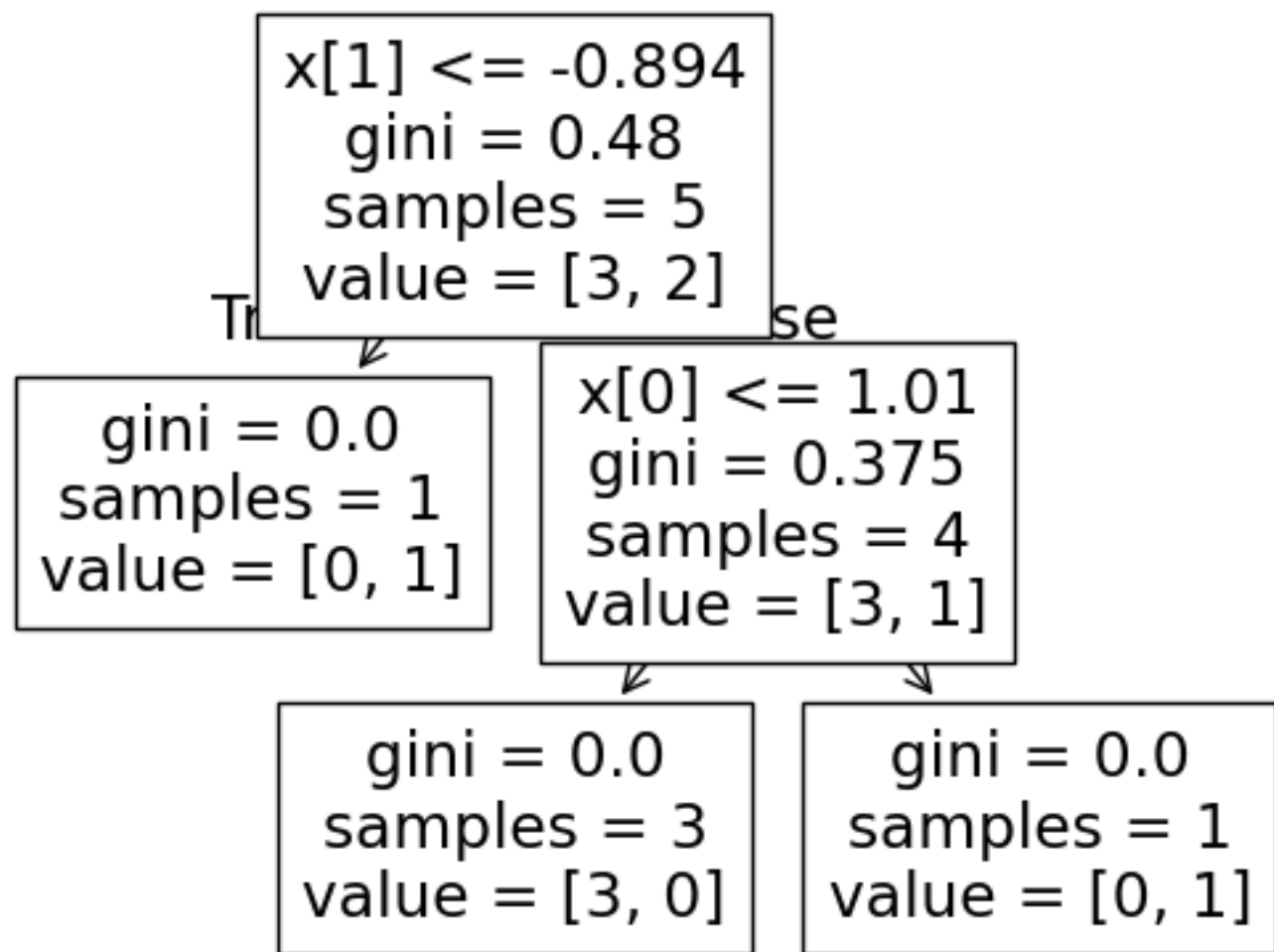
containing large % of unique values)

Linear Model Coefficients

In models like linear or logistic regression the magnitude (and sign) of coefficients indicates the importance and direction of influence for each feature.

Tree Based (Impurity Based)

$$n_i = \frac{N-t}{N} \left[\text{impurity} - \left(\frac{N-t-r}{N-t} \times \text{right impurity} \right) - \left(\frac{N-t-l}{N-t} \times \text{Left impurity} \right) \right]$$



$$N_t = \text{Samples} = 5$$

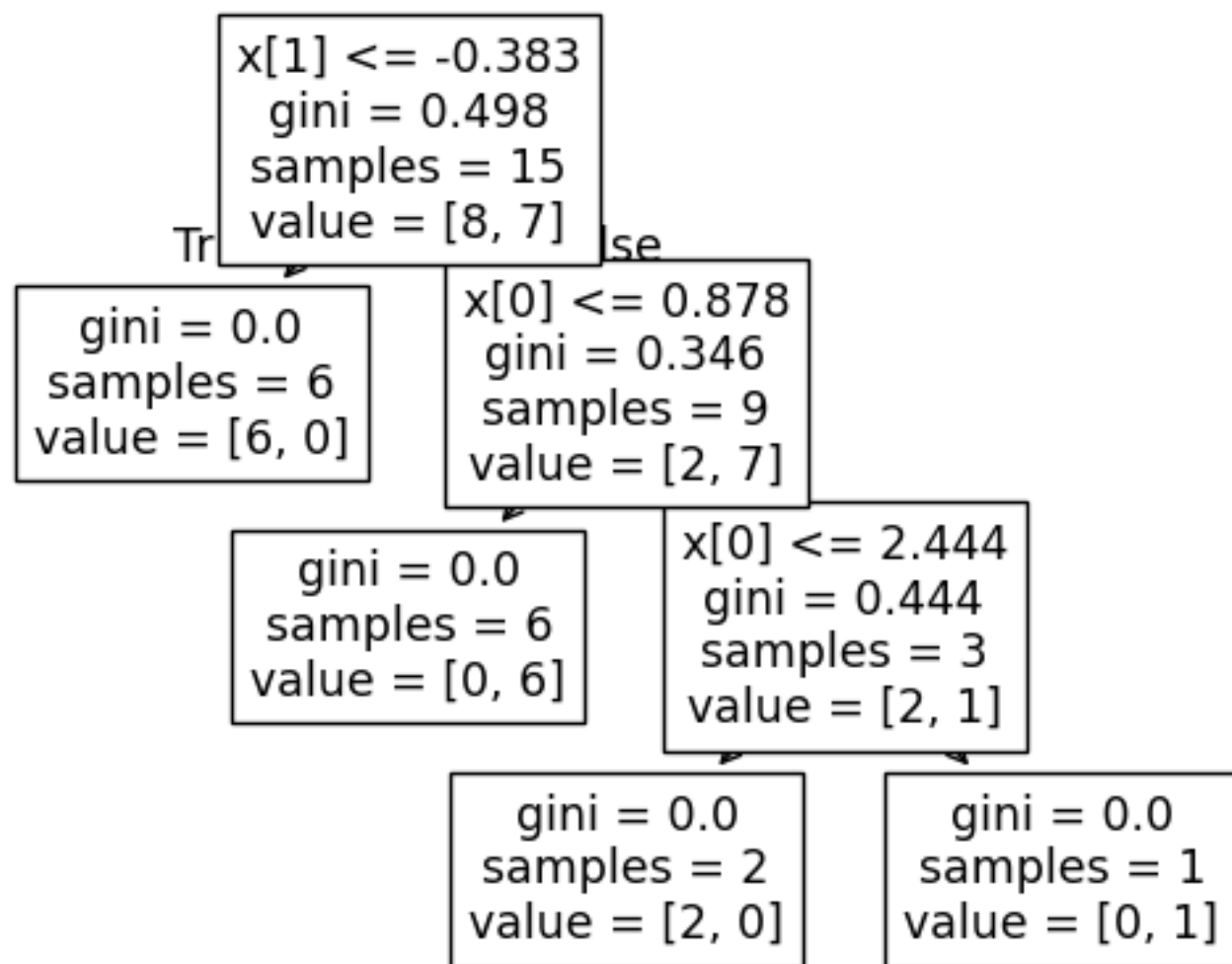
$$n_i = \frac{5}{5} \left[0.48 - \frac{4}{5} \times 0.375 - \frac{1}{5} \times 0 \right]$$

$$n_i = 0.18 \Rightarrow \text{Node Importance} = x$$

$$y = 0.48 - (0.8 \times 0.375) = 0.18$$

$$\begin{array}{l} 0^{\text{th}} \text{ column} \\ \text{importance} \end{array} = \frac{x}{x+y} = 0.625$$

$$1^{\text{st}} = \frac{y}{x+y} = 0.375$$



$$0^{\text{th}} \text{ node} = \frac{15}{15} \left[0.49 - \frac{9}{15} \times 0.34 - 0 \right]$$

$$= 0.29$$

$$1^{\text{st}} \text{ node} = \frac{9}{15} \left[0.34 - \frac{3}{15} \times 0.44 - 0 \right]$$

$$= 0.11$$

$$2^{\text{nd}} \text{ node} = \frac{3}{15} \times 0.44 = 0.08$$

$$FI[0] = \frac{0.11 + 0.08}{0.29 + 0.11 + 0.08} = 0.39$$

$$FI[1] = \frac{0.29}{0.48} = 0.61$$